Informe Actividad Simulando routers BGP

Integrantes

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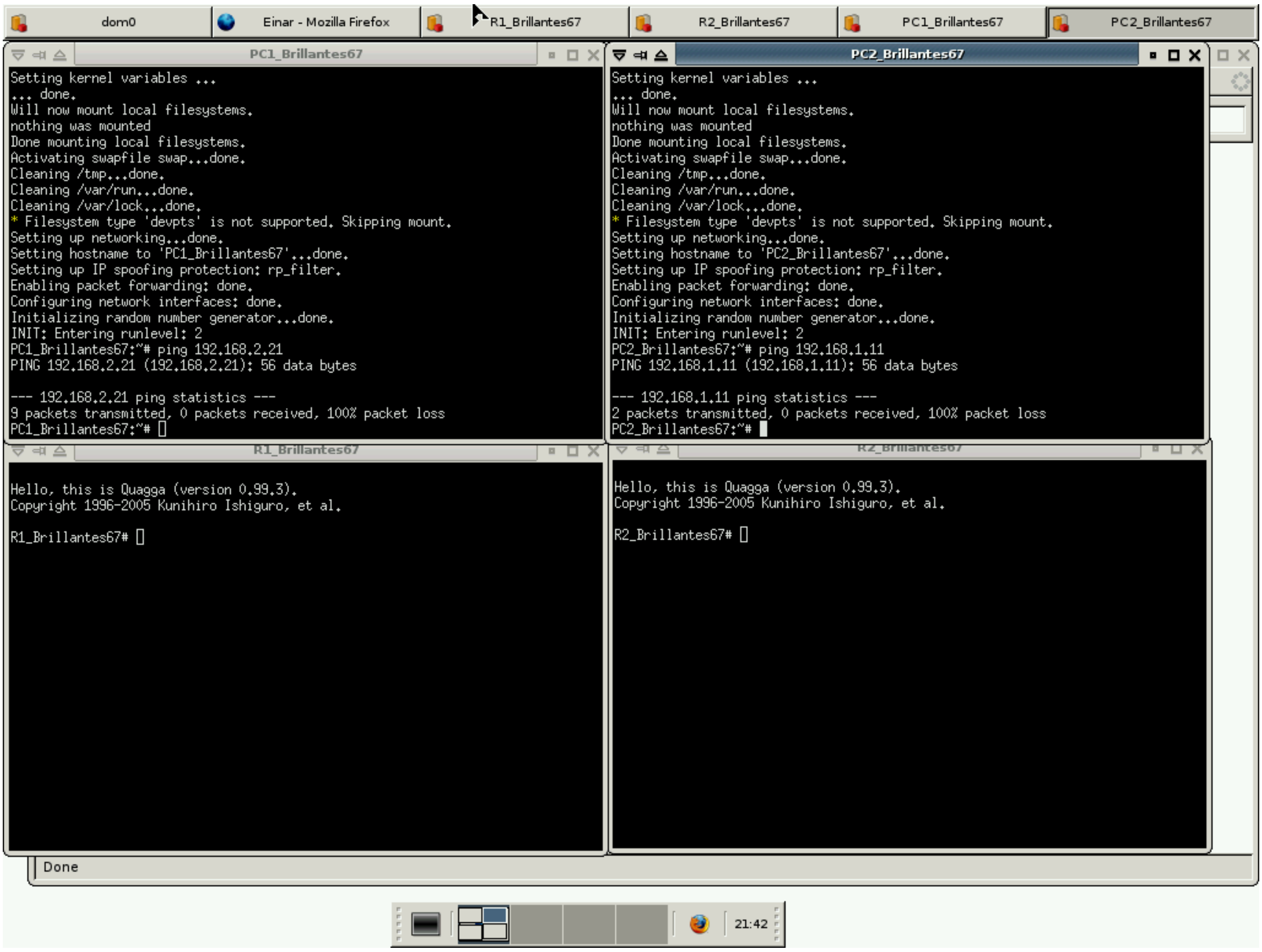
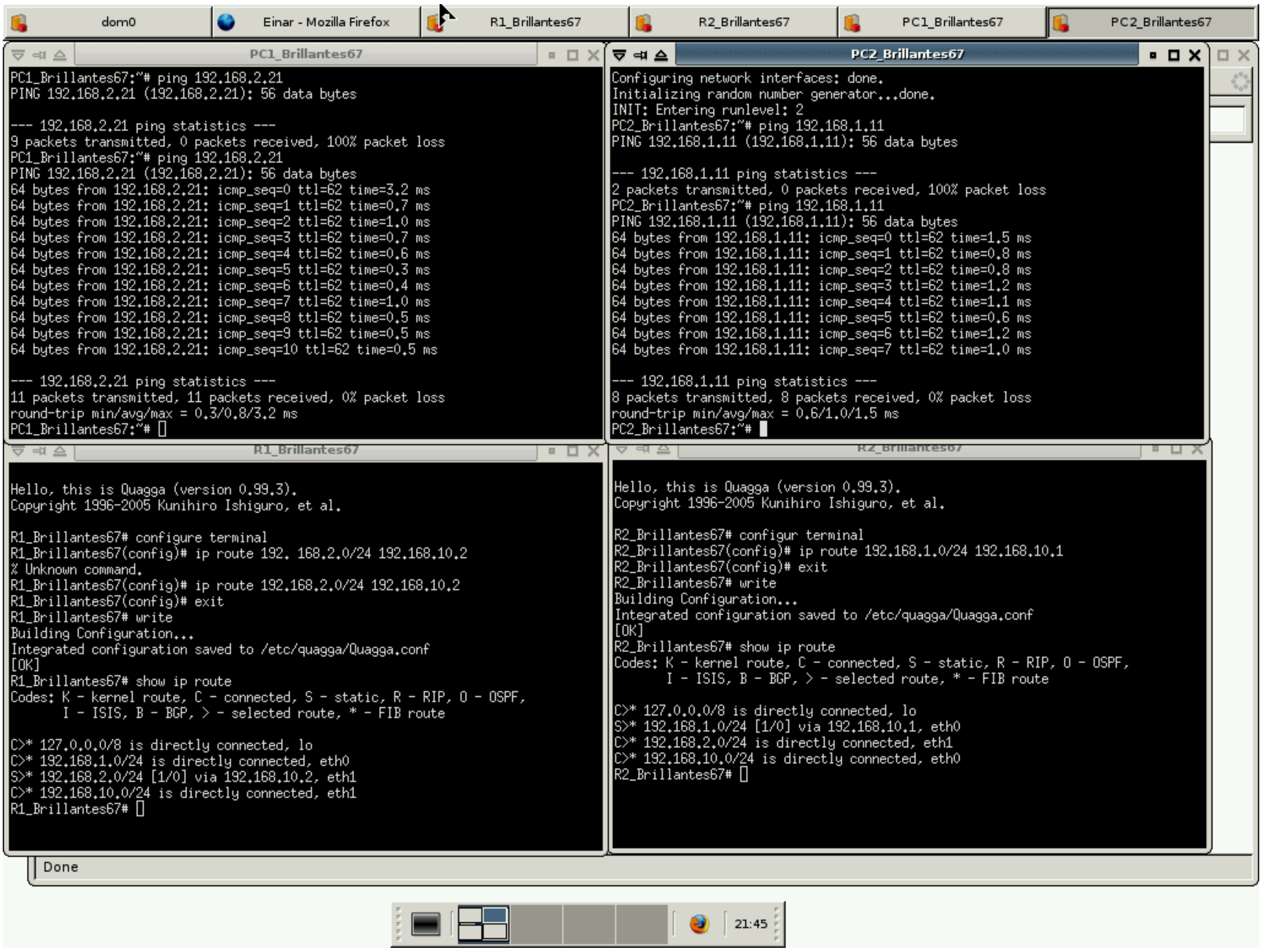
Equipo

Brillantes_67

2-Ejemplo Rutas

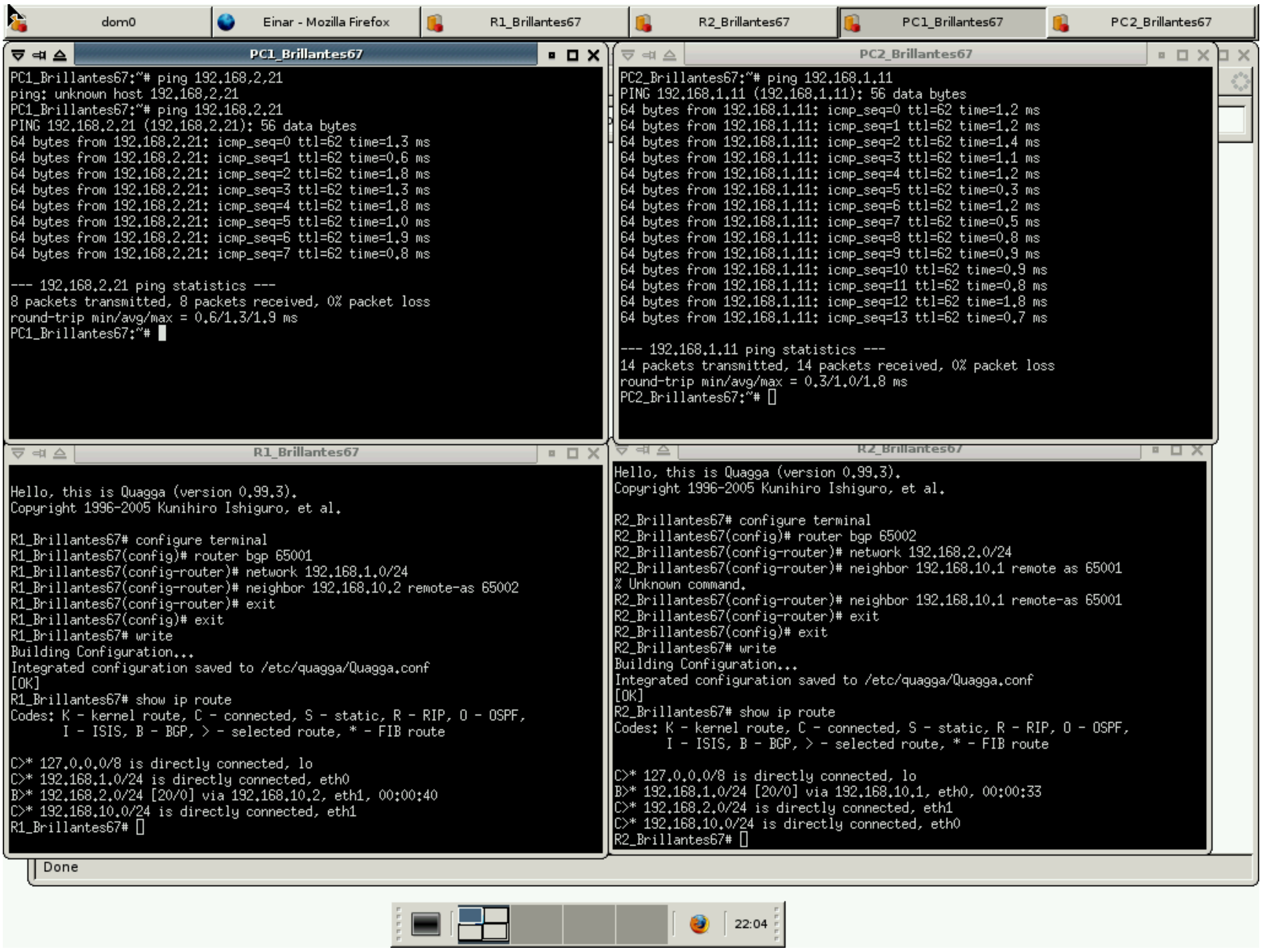
La verdad es que el primer ejemplo esta muy bien explicado por lo que es muy directo aplicar y entender lo ocurrido, al poder ver con ip show route se entiende mejor el funcionamiento. Ademas es interesante configurar un router ya que podemos manejar como se comporta el viaje de informacion de forma mas personalizada. En la primera imagen se ve el no funcionamiento de ping de PC1 a PC2 y en la segunda la adicion de rutas y el funcionamiento de ping

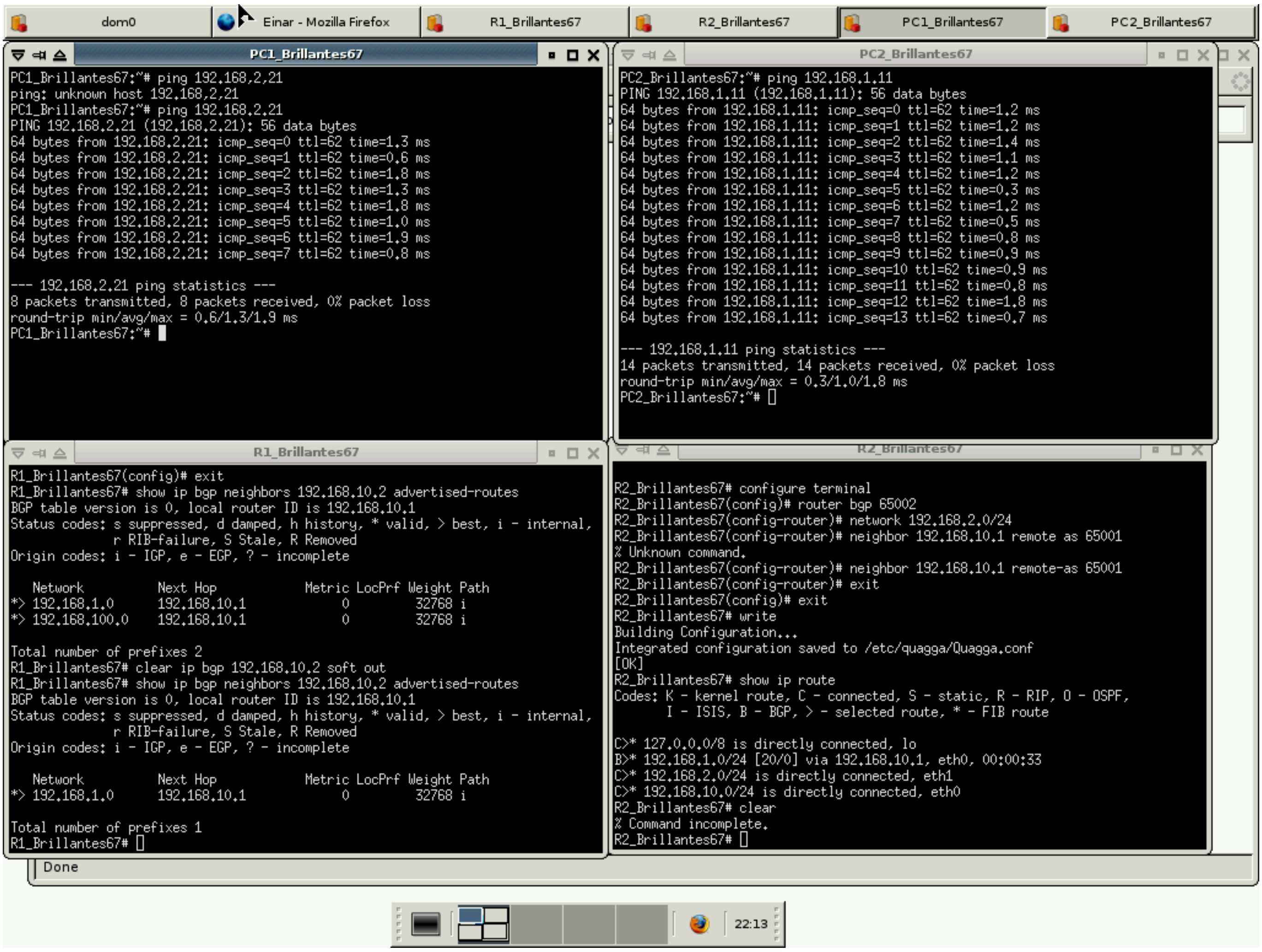

dom0
Einar - Mozilla Firefox
R1_Brillantes67
R2_Brillantes67
PC1_Brillantes67
PC2_Brillantes67
PC1_Brillantes67
Setting kernel variables ...
... done.
Will now mount local filesystems.
nothing was mounted
Done mounting local filesystems.
Activating swapfile swap...done.
Cleaning /tmp...done.
Cleaning /var/run...done.
Cleaning /var/lock...done.
* Filesystem type 'devpts' is not supported. Skipping mount.
Setting up networking...done.
Setting hostname to 'PC1_Brillantes67'...done.
Setting up IP spoofing protection: rp_filter.
Enabling packet forwarding: done.
Configuring network interfaces: done.
Initializing random number generator...done.
INIT: Entering runlevel: 2
PC1_Brillantes67:~# ping 192.168.2.21
PING 192.168.2.21 (192.168.2.21): 56 data bytes
--- 192.168.2.21 ping statistics ---
9 packets transmitted, 0 packets received, 100% packet loss
PC1_Brillantes67:~#
PC2_Brillantes67
Setting kernel variables ...
... done.
Will now mount local filesystems.
nothing was mounted
Done mounting local filesystems.
Activating swapfile swap...done.
Cleaning /tmp...done.
Cleaning /var/run...done.
Cleaning /var/lock...done.
* Filesystem type 'devpts' is not supported. Skipping mount.
Setting up networking...done.
Setting hostname to 'PC2_Brillantes67'...done.
Setting up IP spoofing protection: rp_filter.
Enabling packet forwarding: done.
Configuring network interfaces: done.
Initializing random number generator...done.
INIT: Entering runlevel: 2
PC2_Brillantes67:~# ping 192.168.1.11
PING 192.168.1.11 (192.168.1.11): 56 data bytes
--- 192.168.1.11 ping statistics ---
2 packets transmitted, 0 packets received, 100% packet loss
PC2_Brillantes67:~#
R1_Brillantes67
Hello, this is Quagga (version 0.99.3).
Copyright 1996-2005 Kunihiro Ishiguro, et al.
R1_Brillantes67#
R2_Brillantes67
Hello, this is Quagga (version 0.99.3).
Copyright 1996-2005 Kunihiro Ishiguro, et al.
R2_Brillantes67#
Done
21:42

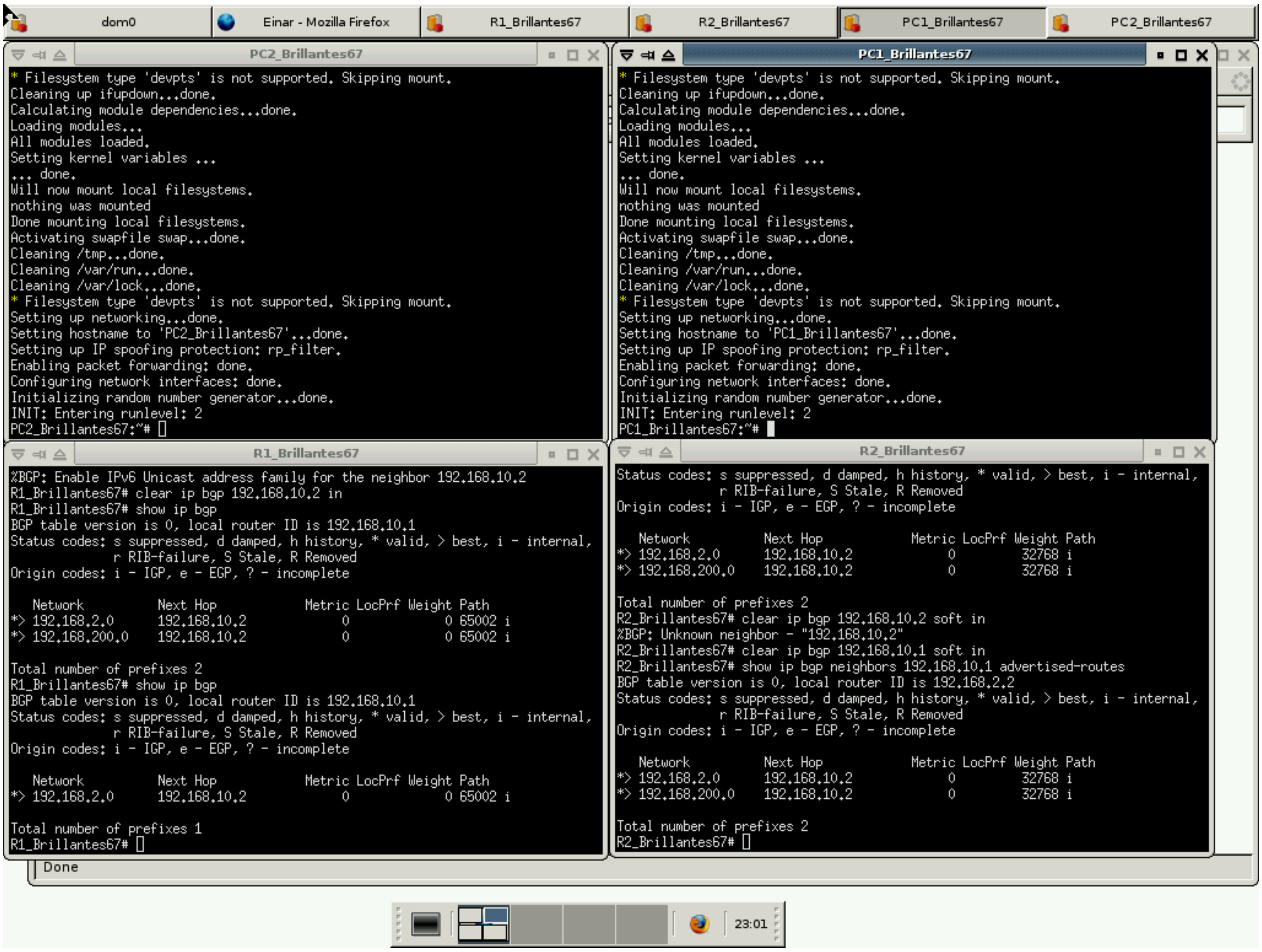


3-Ejemplo BGP

El uso de los prefijos y filtros fue lo mas complejo, con las explicaciones en detalle del eol y realizando la actividad fue mas facil entenderlo, ademas es interesante configurar un roouter bgp porque permite una conexion directa como se vio en la primera parte de esta actividad y ademas permite comunicacion con servicios autonomos externos. Finalmente en la ultima parte hubo que hacer clear en ambas consolas para que funcionara aun siguiendo los pasos de la actividad lo que fue extraño ya que no se espcecificaba pero funciono. En la primera imagen se ve el funcionamiento de los routers bgp sin filtros, en la segunda como funciona el primer filtro y en la 3era como funciona el 2do filtro.

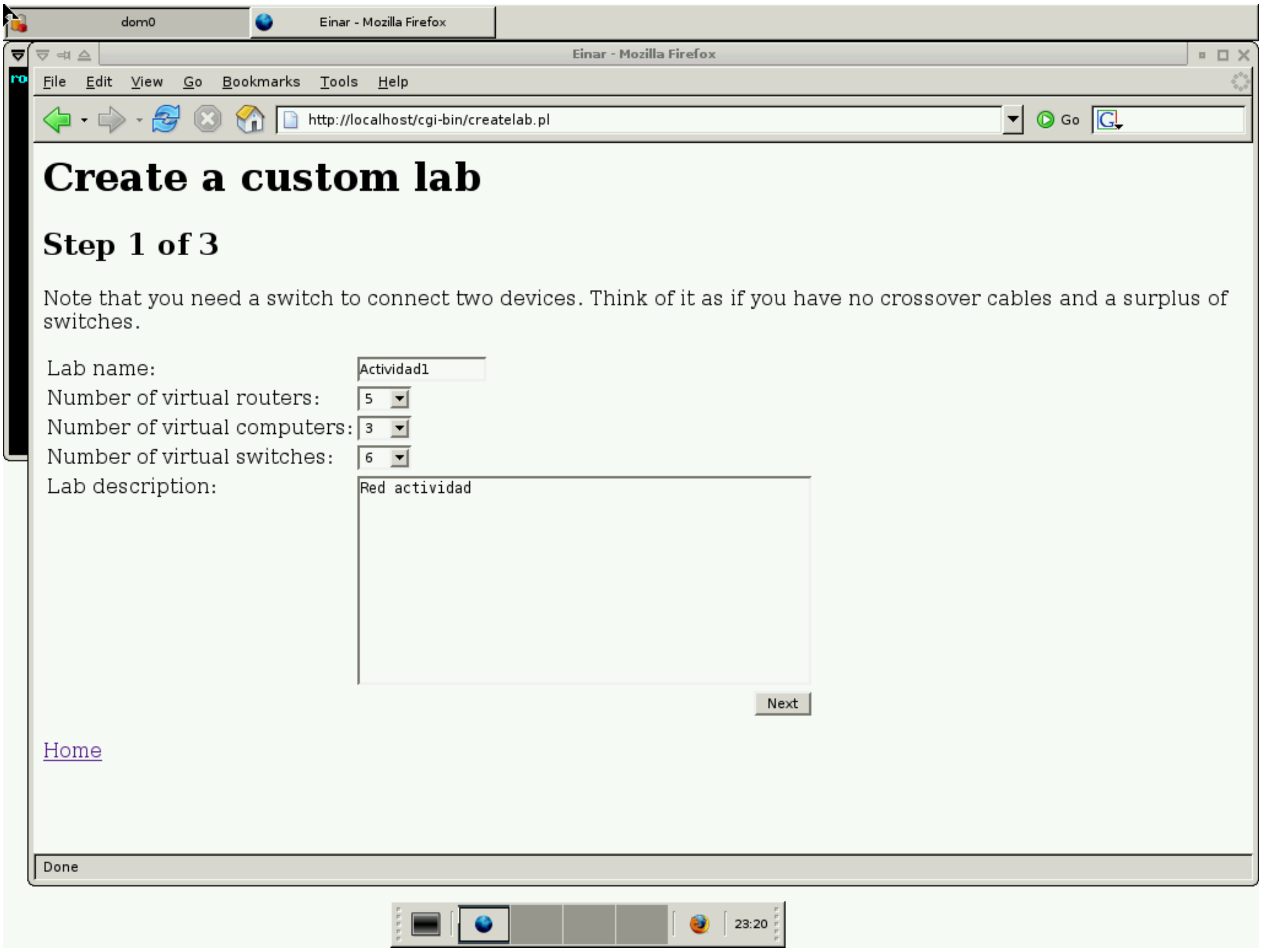

dom0
Einar - Mozilla Firefox
R1_Brillantes67
R2_Brillantes67
PC1_Brillantes67
PC2_Brillantes67
PC1_Brillantes67
PC1_Brillantes67:~# ping 192.168.2.21
ping: unknown host 192.168.2.21
PC1_Brillantes67:~# ping 192.168.2.21
PING 192.168.2.21 (192.168.2.21): 56 data bytes
64 bytes from 192.168.2.21: icmp_seq=0 ttl=62 time=1.3 ms
64 bytes from 192.168.2.21: icmp_seq=1 ttl=62 time=0.6 ms
64 bytes from 192.168.2.21: icmp_seq=2 ttl=62 time=1.8 ms
64 bytes from 192.168.2.21: icmp_seq=3 ttl=62 time=1.3 ms
64 bytes from 192.168.2.21: icmp_seq=4 ttl=62 time=1.8 ms
64 bytes from 192.168.2.21: icmp_seq=5 ttl=62 time=1.0 ms
64 bytes from 192.168.2.21: icmp_seq=6 ttl=62 time=1.9 ms
64 bytes from 192.168.2.21: icmp_seq=7 ttl=62 time=0.8 ms
--- 192.168.2.21 ping statistics ---
8 packets transmitted, 8 packets received, 0% packet loss
round-trip min/avg/max = 0.6/1.3/1.9 ms
PC1_Brillantes67:~#
PC2_Brillantes67
PC2_Brillantes67:~# ping 192.168.1.11
PING 192.168.1.11 (192.168.1.11): 56 data bytes
64 bytes from 192.168.1.11: icmp_seq=0 ttl=62 time=1.2 ms
64 bytes from 192.168.1.11: icmp_seq=1 ttl=62 time=1.2 ms
64 bytes from 192.168.1.11: icmp_seq=2 ttl=62 time=1.4 ms
64 bytes from 192.168.1.11: icmp_seq=3 ttl=62 time=1.1 ms
64 bytes from 192.168.1.11: icmp_seq=4 ttl=62 time=1.2 ms
64 bytes from 192.168.1.11: icmp_seq=5 ttl=62 time=0.3 ms
64 bytes from 192.168.1.11: icmp_seq=6 ttl=62 time=1.2 ms
64 bytes from 192.168.1.11: icmp_seq=7 ttl=62 time=0.5 ms
64 bytes from 192.168.1.11: icmp_seq=8 ttl=62 time=0.8 ms
64 bytes from 192.168.1.11: icmp_seq=9 ttl=62 time=0.9 ms
64 bytes from 192.168.1.11: icmp_seq=10 ttl=62 time=0.9 ms
64 bytes from 192.168.1.11: icmp_seq=11 ttl=62 time=0.8 ms
64 bytes from 192.168.1.11: icmp_seq=12 ttl=62 time=1.8 ms
64 bytes from 192.168.1.11: icmp_seq=13 ttl=62 time=0.7 ms
--- 192.168.1.11 ping statistics ---
14 packets transmitted, 14 packets received, 0% packet loss
round-trip min/avg/max = 0.3/1.0/1.8 ms
PC2_Brillantes67:~#
R1_Brillantes67
Hello, this is Quagga (version 0.99.3).
Copyright 1996-2005 Kunihiro Ishiguro, et al.
R1_Brillantes67# configure terminal
R1_Brillantes67(config)# router bgp 65001
R1_Brillantes67(config-router)# network 192.168.1.0/24
R1_Brillantes67(config-router)# neighbor 192.168.10.2 remote-as 65002
R1_Brillantes67(config-router)# exit
R1_Brillantes67(config)# exit
R1_Brillantes67# write
Building Configuration...
Integrated configuration saved to /etc/quagga/Quagga.conf
[OK]
R1_Brillantes67# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP, O - OSPF,
I - ISIS, B - BGP, > - selected route, * - FIB route
C>* 127.0.0.0/8 is directly connected, lo
C>* 192.168.1.0/24 is directly connected, eth0
B>* 192.168.2.0/24 [20/0] via 192.168.10.2, eth1, 00:00:40
C>* 192.168.10.0/24 is directly connected, eth1
R1_Brillantes67#
R2_Brillantes67
Hello, this is Quagga (version 0.99.3).
Copyright 1996-2005 Kunihiro Ishiguro, et al.
R2_Brillantes67# configure terminal
R2_Brillantes67(config)# router bgp 65002
R2_Brillantes67(config-router)# network 192.168.2.0/24
R2_Brillantes67(config-router)# neighbor 192.168.10.1 remote as 65001
% Unknown command.
R2_Brillantes67(config-router)# neighbor 192.168.10.1 remote-as 65001
R2_Brillantes67(config-router)# exit
R2_Brillantes67(config)# exit
R2_Brillantes67# write
Building Configuration...
Integrated configuration saved to /etc/quagga/Quagga.conf
[OK]
R2_Brillantes67# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP, O - OSPF,
I - ISIS, B - BGP, > - selected route, * - FIB route
C>* 127.0.0.0/8 is directly connected, lo
B>* 192.168.1.0/24 [20/0] via 192.168.10.1, eth0, 00:00:33
C>* 192.168.2.0/24 is directly connected, eth1
C>* 192.168.10.0/24 is directly connected, eth0
R2_Brillantes67#
Done
22:04

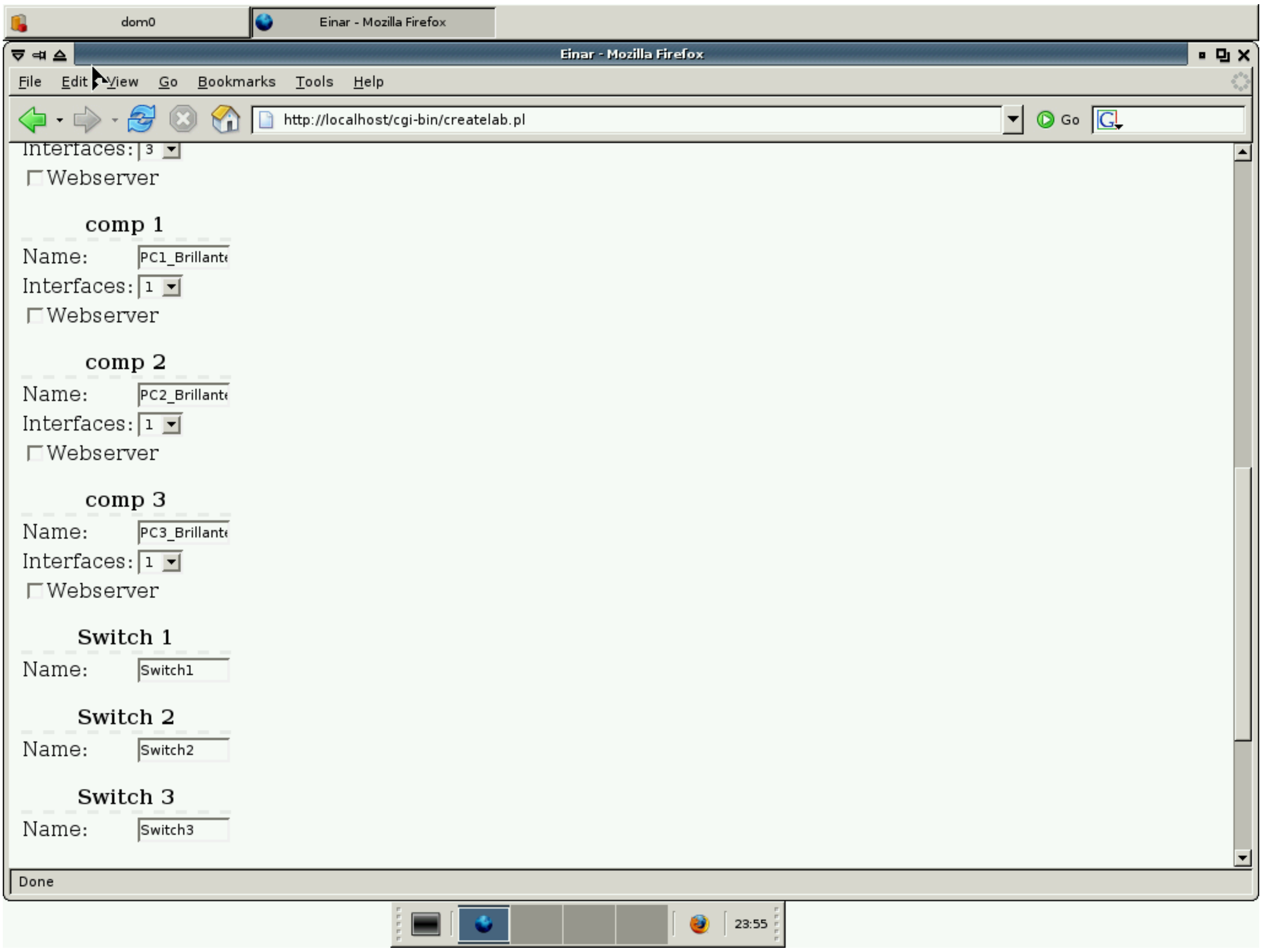

dom0
Einar - Mozilla Firefox
R1_Brillantes67
R2_Brillantes67
PC1_Brillantes67
PC2_Brillantes67
PC1_Brillantes67
PC1_Brillantes67:~# ping 192.168.2.21
ping: unknown host 192.168.2.21
PC1_Brillantes67:~# ping 192.168.2.21
PING 192.168.2.21 (192.168.2.21): 56 data bytes
64 bytes from 192.168.2.21: icmp_seq=0 ttl=62 time=1.3 ms
64 bytes from 192.168.2.21: icmp_seq=1 ttl=62 time=0.6 ms
64 bytes from 192.168.2.21: icmp_seq=2 ttl=62 time=1.8 ms
64 bytes from 192.168.2.21: icmp_seq=3 ttl=62 time=1.3 ms
64 bytes from 192.168.2.21: icmp_seq=4 ttl=62 time=1.8 ms
64 bytes from 192.168.2.21: icmp_seq=5 ttl=62 time=1.0 ms
64 bytes from 192.168.2.21: icmp_seq=6 ttl=62 time=1.9 ms
64 bytes from 192.168.2.21: icmp_seq=7 ttl=62 time=0.8 ms
--- 192.168.2.21 ping statistics ---
8 packets transmitted, 8 packets received, 0% packet loss
round-trip min/avg/max = 0.6/1.3/1.9 ms
PC1_Brillantes67:~#
PC2_Brillantes67
PC2_Brillantes67:~# ping 192.168.1.11
PING 192.168.1.11 (192.168.1.11): 56 data bytes
64 bytes from 192.168.1.11: icmp_seq=0 ttl=62 time=1.2 ms
64 bytes from 192.168.1.11: icmp_seq=1 ttl=62 time=1.2 ms
64 bytes from 192.168.1.11: icmp_seq=2 ttl=62 time=1.4 ms
64 bytes from 192.168.1.11: icmp_seq=3 ttl=62 time=1.1 ms
64 bytes from 192.168.1.11: icmp_seq=4 ttl=62 time=1.2 ms
64 bytes from 192.168.1.11: icmp_seq=5 ttl=62 time=0.3 ms
64 bytes from 192.168.1.11: icmp_seq=6 ttl=62 time=1.2 ms
64 bytes from 192.168.1.11: icmp_seq=7 ttl=62 time=0.5 ms
64 bytes from 192.168.1.11: icmp_seq=8 ttl=62 time=0.8 ms
64 bytes from 192.168.1.11: icmp_seq=9 ttl=62 time=0.9 ms
64 bytes from 192.168.1.11: icmp_seq=10 ttl=62 time=0.9 ms
64 bytes from 192.168.1.11: icmp_seq=11 ttl=62 time=0.8 ms
64 bytes from 192.168.1.11: icmp_seq=12 ttl=62 time=1.8 ms
64 bytes from 192.168.1.11: icmp_seq=13 ttl=62 time=0.7 ms
--- 192.168.1.11 ping statistics ---
14 packets transmitted, 14 packets received, 0% packet loss
round-trip min/avg/max = 0.3/1.0/1.8 ms
PC2_Brillantes67:~#
R1_Brillantes67
R1_Brillantes67(config)# exit
R1_Brillantes67# show ip bgp neighbors 192.168.10.2 advertised-routes
BGP table version is 0, local router ID is 192.168.10.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
Network Next Hop Metric LocPrf Weight Path
*> 192.168.1.0 192.168.10.1 0 32768 i
*> 192.168.100.0 192.168.10.1 0 32768 i
Total number of prefixes 2
R1_Brillantes67# clear ip bgp 192.168.10.2 soft out
R1_Brillantes67# show ip bgp neighbors 192.168.10.2 advertised-routes
BGP table version is 0, local router ID is 192.168.10.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
Network Next Hop Metric LocPrf Weight Path
*> 192.168.1.0 192.168.10.1 0 32768 i
Total number of prefixes 1
R1_Brillantes67#
R2_Brillantes67
R2_Brillantes67# configure terminal
R2_Brillantes67(config)# router bgp 65002
R2_Brillantes67(config-router)# network 192.168.2.0/24
R2_Brillantes67(config-router)# neighbor 192.168.10.1 remote as 65001
% Unknown command.
R2_Brillantes67(config-router)# neighbor 192.168.10.1 remote-as 65001
R2_Brillantes67(config-router)# exit
R2_Brillantes67(config)# exit
R2_Brillantes67# write
Building Configuration...
Integrated configuration saved to /etc/quagga/Quagga.conf
[OK]
R2_Brillantes67# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP, O - OSPF,
I - ISIS, B - BGP, > - selected route, * - FIB route
C>* 127.0.0.0/8 is directly connected, lo
B>* 192.168.1.0/24 [20/0] via 192.168.10.1, eth0, 00:00:33
C>* 192.168.2.0/24 is directly connected, eth1
C>* 192.168.10.0/24 is directly connected, eth0
R2_Brillantes67# clear
% Command incomplete.
R2_Brillantes67#
Done
22:13

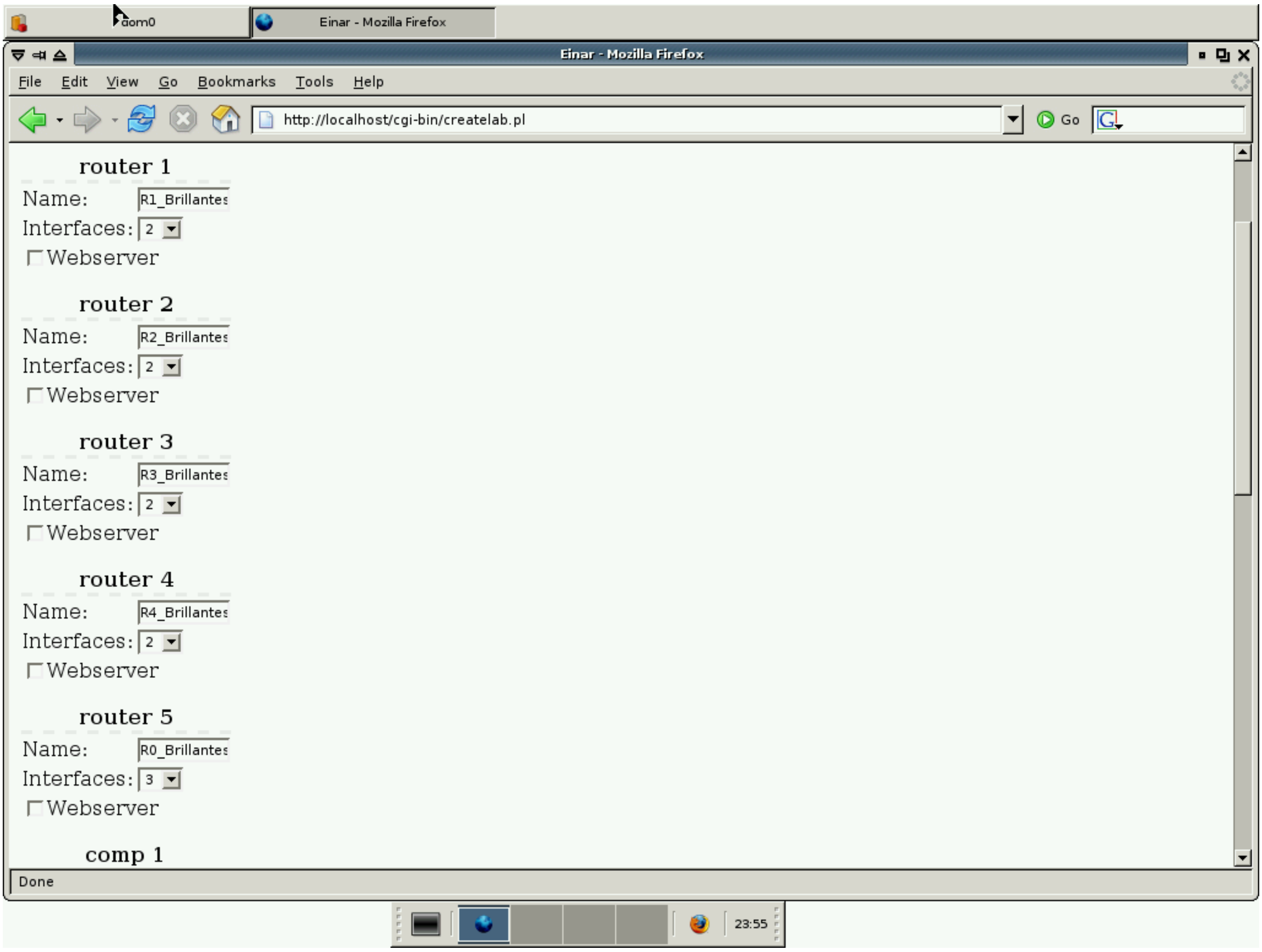


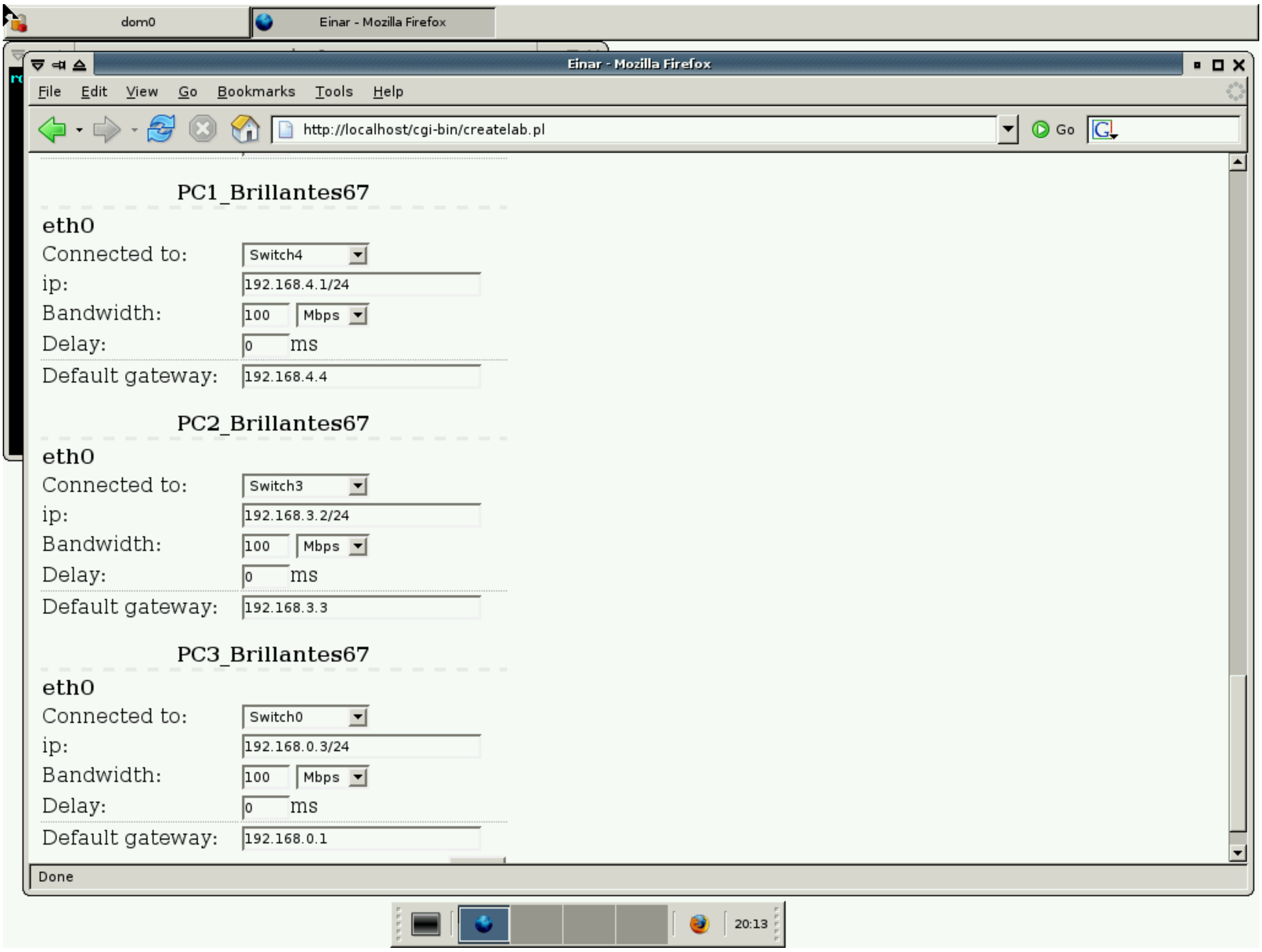
4-Configuracion Einar

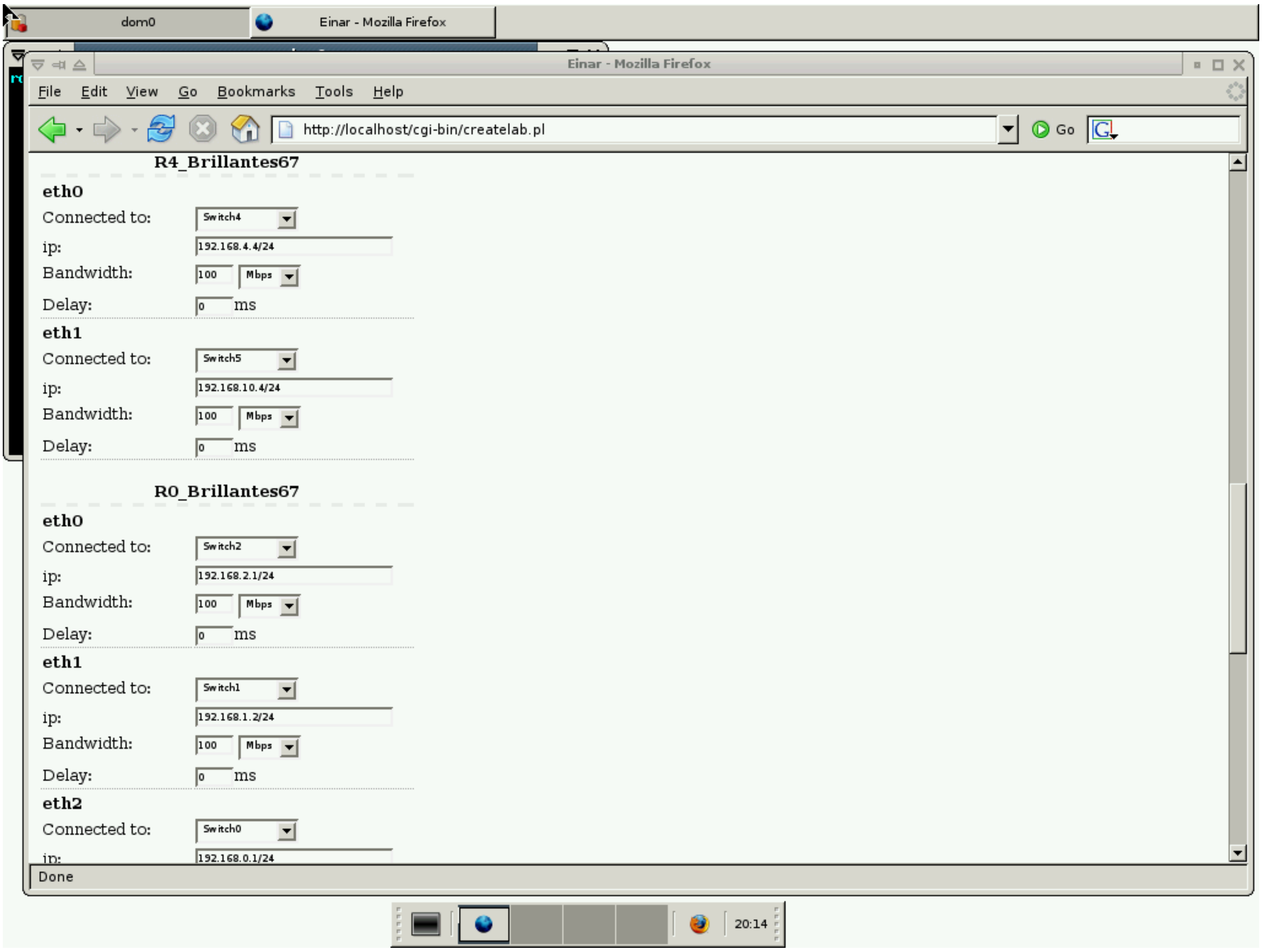
Dejare imagenes de como se configuran cada componente relevante para el laboratorio. La primera imagen es cuantos elementos hay de cada uno, luego se muestran los nombres y cantidad de interfaces de routers y pc's. Finalmente se ve la configuracion con ip's, switches y gateways para pc's.

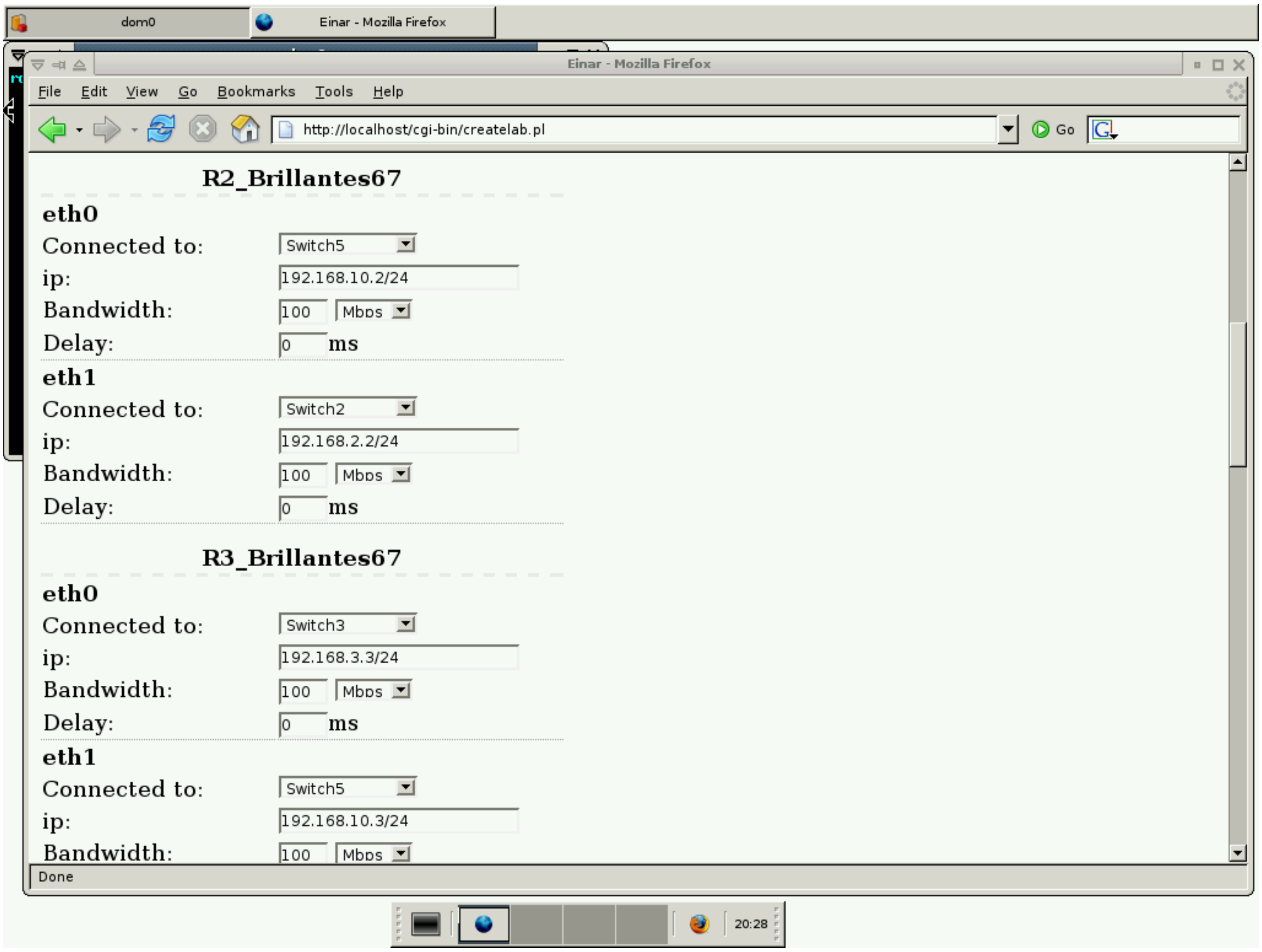

dom0
Einar - Mozilla Firefox
File Edit View Go Bookmarks Tools Help
http://localhost/cgi-bin/createlab.pl
Go
Create a custom lab
Step 1 of 3
Note that you need a switch to connect two devices. Think of it as if you have no crossover cables and a surplus of switches.
Lab name: Actividad1
Number of virtual routers: 5
Number of virtual computers: 3
Number of virtual switches: 6
Lab description: Red actividad
Next
Home
Done
23:20

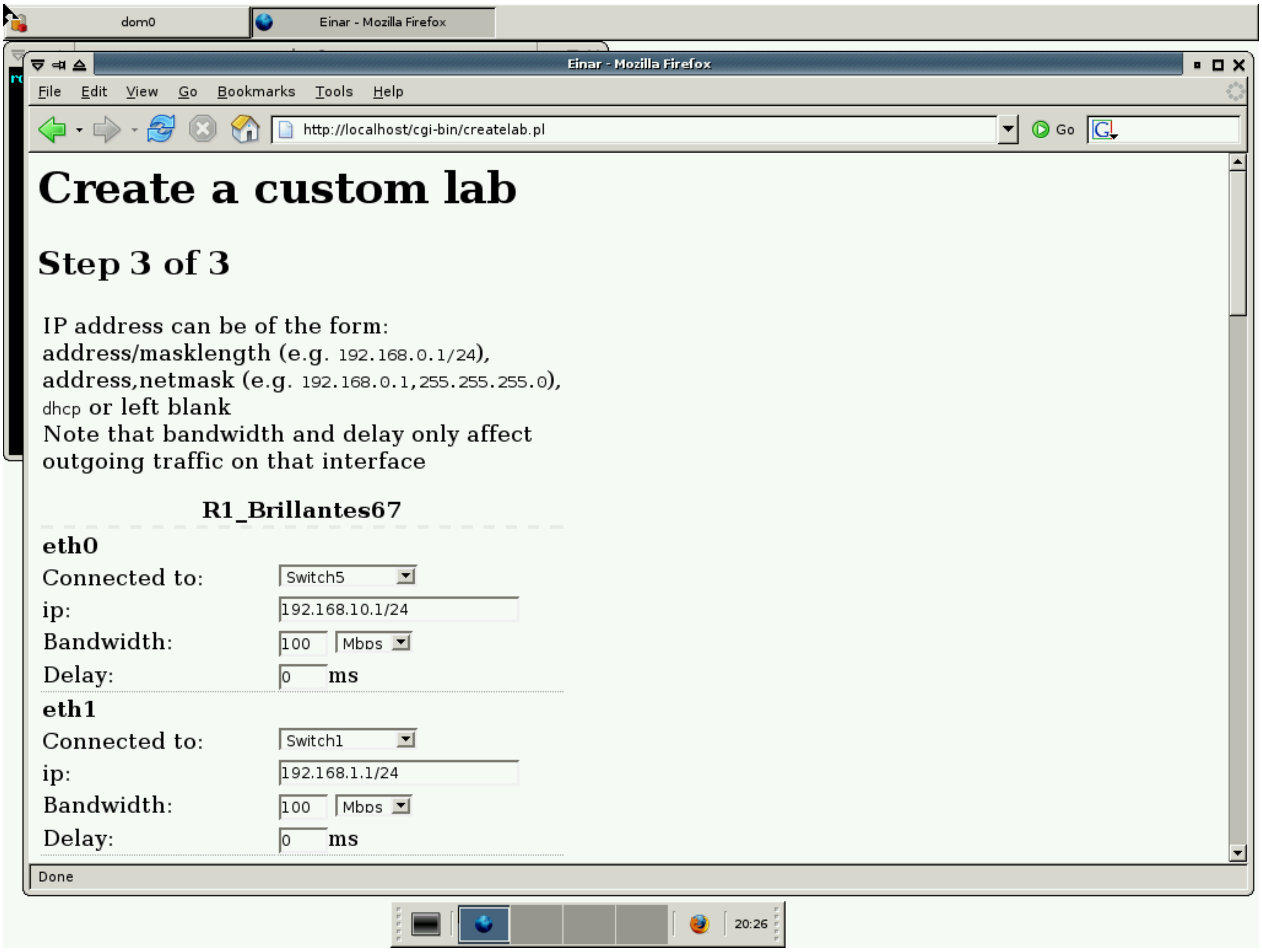

dom0
Einar - Mozilla Firefox
File Edit View Go Bookmarks Tools Help
http://localhost/cgi-bin/createlab.pl
Go
Interfaces: 3
Webserver
comp 1
Name: PC1_Brillant
Interfaces: 1
Webserver
comp 2
Name: PC2_Brillant
Interfaces: 1
Webserver
comp 3
Name: PC3_Brillant
Interfaces: 1
Webserver
Switch 1
Name: Switch1
Switch 2
Name: Switch2
Switch 3
Name: Switch3
Done
23:55


dom0
Einar - Mozilla Firefox
Einar - Mozilla Firefox
File Edit View Go Bookmarks Tools Help
http://localhost/cgi-bin/createlab.pl
Go
router 1
Name: R1_Brillantes
Interfaces: 2
Webserver
router 2
Name: R2_Brillantes
Interfaces: 2
Webserver
router 3
Name: R3_Brillantes
Interfaces: 2
Webserver
router 4
Name: R4_Brillantes
Interfaces: 2
Webserver
router 5
Name: R0_Brillantes
Interfaces: 3
Webserver
comp 1
Done
23:55


dom0
Einar - Mozilla Firefox
Einar - Mozilla Firefox
File Edit View Go Bookmarks Tools Help
http://localhost/cgi-bin/createlab.pl
Go
PC1_Brillantes67
eth0
Connected to: Switch4
ip: 192.168.4.1/24
Bandwidth: 100 Mbps
Delay: 0 ms
Default gateway: 192.168.4.4
PC2_Brillantes67
eth0
Connected to: Switch3
ip: 192.168.3.2/24
Bandwidth: 100 Mbps
Delay: 0 ms
Default gateway: 192.168.3.3
PC3_Brillantes67
eth0
Connected to: Switch0
ip: 192.168.0.3/24
Bandwidth: 100 Mbps
Delay: 0 ms
Default gateway: 192.168.0.1
Done
20:13

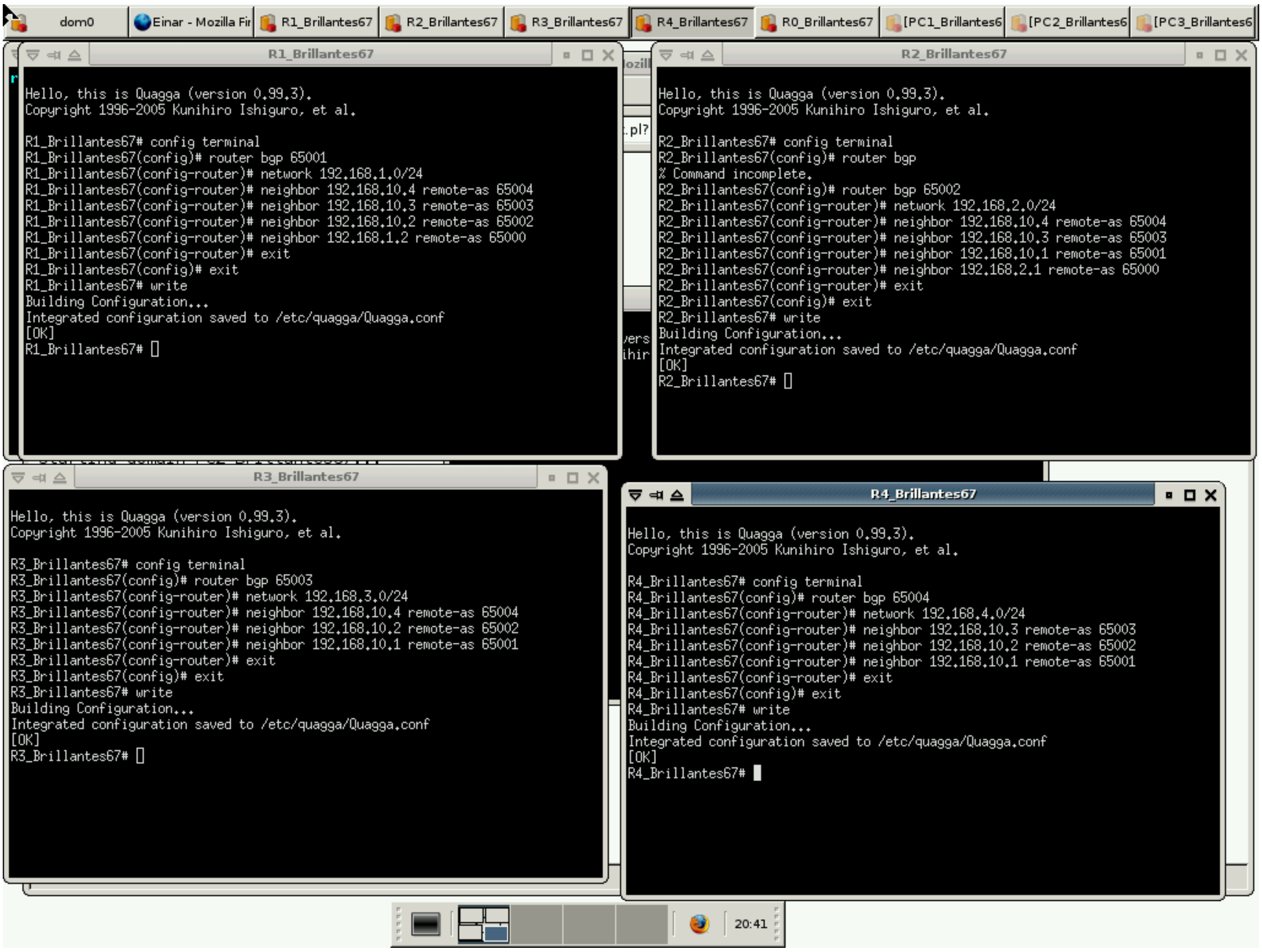

dom0
Einar - Mozilla Firefox
Einar - Mozilla Firefox
File Edit View Go Bookmarks Tools Help
http://localhost/cgi-bin/createlab.pl
Go
R4_Brillantes67
eth0
Connected to: Switch4
ip: 192.168.4.4/24
Bandwidth: 100 Mbps
Delay: 0 ms
eth1
Connected to: Switch5
ip: 192.168.10.4/24
Bandwidth: 100 Mbps
Delay: 0 ms
R0_Brillantes67
eth0
Connected to: Switch2
ip: 192.168.2.1/24
Bandwidth: 100 Mbps
Delay: 0 ms
eth1
Connected to: Switch1
ip: 192.168.1.2/24
Bandwidth: 100 Mbps
Delay: 0 ms
eth2
Connected to: Switch0
ip: 192.168.0.1/24
Done
20:14

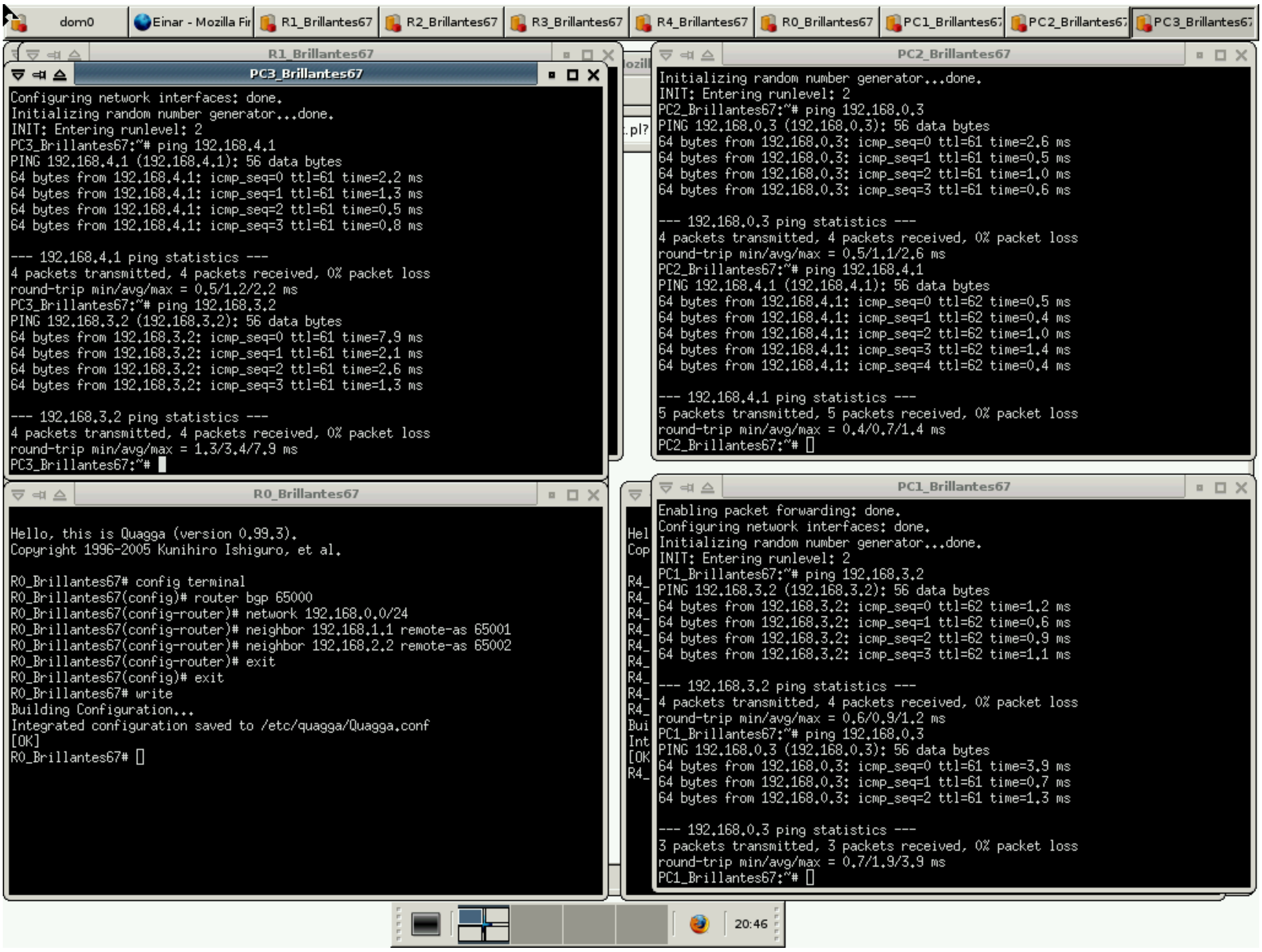

dom0
Einar - Mozilla Firefox
File Edit View Go Bookmarks Tools Help
http://localhost/cgi-bin/createlab.pl
Go
R2_Brillantes67
eth0
Connected to: Switch5
ip: 192.168.10.2/24
Bandwidth: 100 Mbps
Delay: 0 ms
eth1
Connected to: Switch2
ip: 192.168.2.2/24
Bandwidth: 100 Mbps
Delay: 0 ms
R3_Brillantes67
eth0
Connected to: Switch3
ip: 192.168.3.3/24
Bandwidth: 100 Mbps
Delay: 0 ms
eth1
Connected to: Switch5
ip: 192.168.10.3/24
Bandwidth: 100 Mbps
Done
20:28


dom0
Einar - Mozilla Firefox
File Edit View Go Bookmarks Tools Help
http://localhost/cgi-bin/createlab.pl
Go
Create a custom lab
Step 3 of 3
IP address can be of the form:
address/masklength (e.g. 192.168.0.1/24),
address,netmask (e.g. 192.168.0.1,255.255.255.0),
dhcp or left blank
Note that bandwidth and delay only affect
outgoing traffic on that interface
R1_Brillantes67
eth0
Connected to: Switch5
ip: 192.168.10.1/24
Bandwidth: 100 Mbps
Delay: 0 ms
eth1
Connected to: Switch1
ip: 192.168.1.1/24
Bandwidth: 100 Mbps
Delay: 0 ms
Done
20:26

5-Ruteo BGP

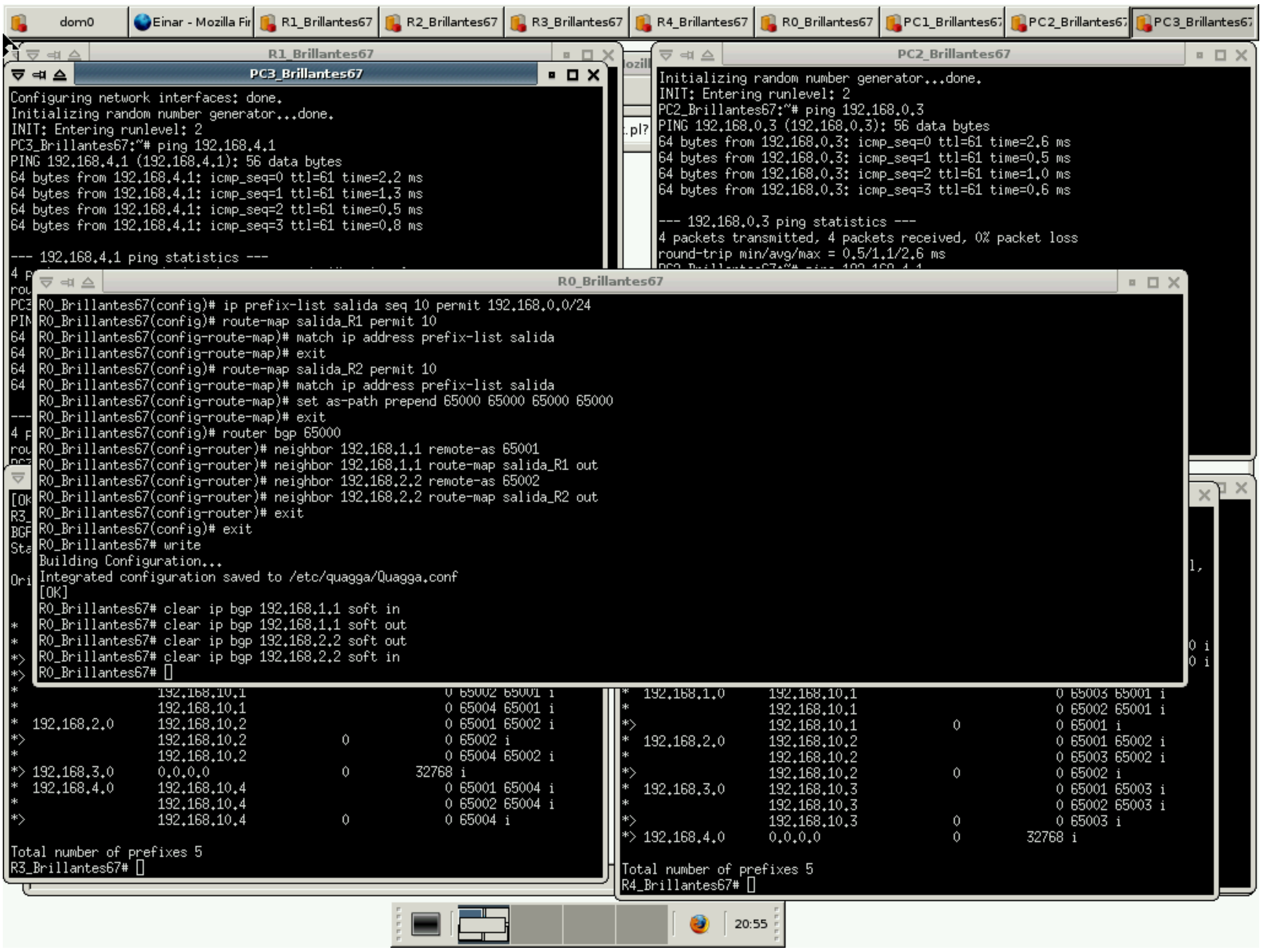
Se ve la config terminal de cada router



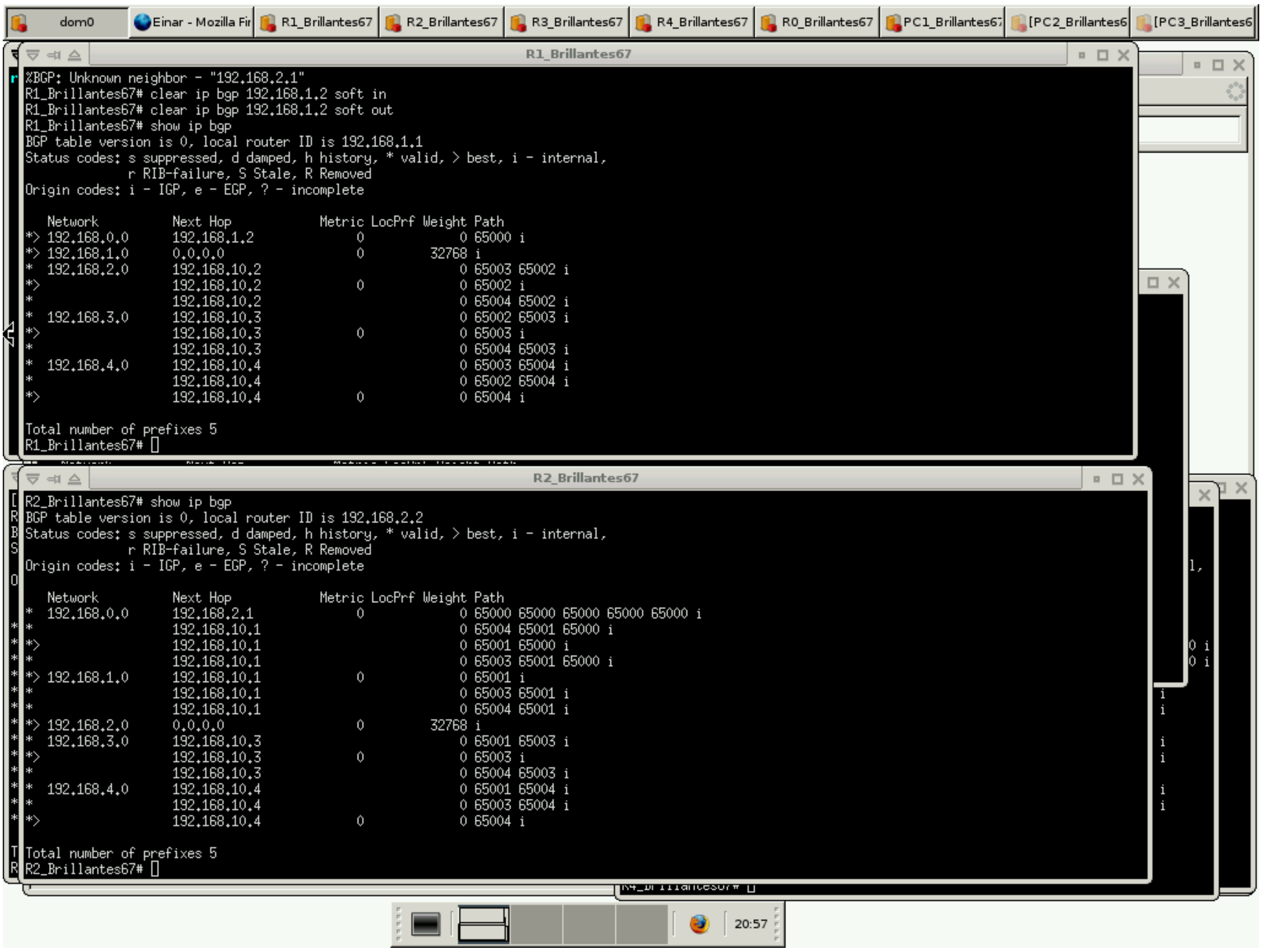


6-Configuracion R0

Empezamos por entender que creamos una lista de prefijos que solo incluye a las ip 192.168.0.0/24, o sea el PC3 y le permite salir. Luego en R1 lo dejamos con esta lista y no hacemos nada mas pero a R2 agregamos 4 saltos ciclicos en R0 antes de seguir por lo que se demorara 4 pasos mas. Por esto mismo la ruta elegida de R2 para llegar a R0 sera por R1 por raro que parezca. Se puede ver en las siguientes imagenes ademas marcado por el >.



PC3_Brillantes67
Configuring network interfaces: done.
Initializing random number generator...done.
INIT: Entering runlevel: 2
PC3_Brillantes67:~# ping 192.168.4.1
PING 192.168.4.1 (192.168.4.1): 56 data bytes
64 bytes from 192.168.4.1: icmp_seq=0 ttl=61 time=2.2 ms
64 bytes from 192.168.4.1: icmp_seq=1 ttl=61 time=1.3 ms
64 bytes from 192.168.4.1: icmp_seq=2 ttl=61 time=0.5 ms
64 bytes from 192.168.4.1: icmp_seq=3 ttl=61 time=0.8 ms
--- 192.168.4.1 ping statistics ---
PC2_Brillantes67
Initializing random number generator...done.
INIT: Entering runlevel: 2
PC2_Brillantes67:~# ping 192.168.0.3
PING 192.168.0.3 (192.168.0.3): 56 data bytes
64 bytes from 192.168.0.3: icmp_seq=0 ttl=61 time=2.6 ms
64 bytes from 192.168.0.3: icmp_seq=1 ttl=61 time=0.5 ms
64 bytes from 192.168.0.3: icmp_seq=2 ttl=61 time=1.0 ms
64 bytes from 192.168.0.3: icmp_seq=3 ttl=61 time=0.6 ms
--- 192.168.0.3 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.5/1.1/2.6 ms
R0_Brillantes67
R0_Brillantes67(config)# ip prefix-list salida seq 10 permit 192.168.0.0/24
R0_Brillantes67(config)# route-map salida_R1 permit 10
R0_Brillantes67(config-route-map)# match ip address prefix-list salida
R0_Brillantes67(config-route-map)# exit
R0_Brillantes67(config)# route-map salida_R2 permit 10
R0_Brillantes67(config-route-map)# match ip address prefix-list salida
R0_Brillantes67(config-route-map)# set as-path prepend 65000 65000 65000 65000
R0_Brillantes67(config-route-map)# exit
R0_Brillantes67(config)# router bgp 65000
R0_Brillantes67(config-router)# neighbor 192.168.1.1 remote-as 65001
R0_Brillantes67(config-router)# neighbor 192.168.1.1 route-map salida_R1 out
R0_Brillantes67(config-router)# neighbor 192.168.2.2 remote-as 65002
R0_Brillantes67(config-router)# neighbor 192.168.2.2 route-map salida_R2 out
R0_Brillantes67(config-router)# exit
R0_Brillantes67(config)# exit
R0_Brillantes67# write
Building Configuration...
Integrated configuration saved to /etc/quagga/Quagga.conf
[OK]
R0_Brillantes67# clear ip bgp 192.168.1.1 soft in
R0_Brillantes67# clear ip bgp 192.168.1.1 soft out
R0_Brillantes67# clear ip bgp 192.168.2.2 soft out
R0_Brillantes67# clear ip bgp 192.168.2.2 soft in
R0_Brillantes67#
* 192.168.2.0 192.168.10.2 0 65001 65002 i
*> 192.168.10.2 0 0 65002 i
* 192.168.10.2 0 65004 65002 i
*> 192.168.3.0 0.0.0.0 0 32768 i
* 192.168.4.0 192.168.10.4 0 65001 65004 i
* 192.168.10.4 0 65002 65004 i
*> 192.168.10.4 0 0 65004 i
Total number of prefixes 5
R3_Brillantes67#
* 192.168.1.0 192.168.10.1 0 65003 65001 i
* 192.168.10.1 0 65002 65001 i
*> 192.168.10.1 0 0 65001 i
* 192.168.2.0 192.168.10.2 0 65001 65002 i
* 192.168.10.2 0 65003 65002 i
*> 192.168.10.2 0 0 65002 i
* 192.168.3.0 192.168.10.3 0 65001 65003 i
* 192.168.10.3 0 65002 65003 i
*> 192.168.10.3 0 0 65003 i
*> 192.168.4.0 0.0.0.0 0 32768 i
Total number of prefixes 5
R4_Brillantes67#



7-Configurando R3 y R4

Los comandos a utilizar son estos.

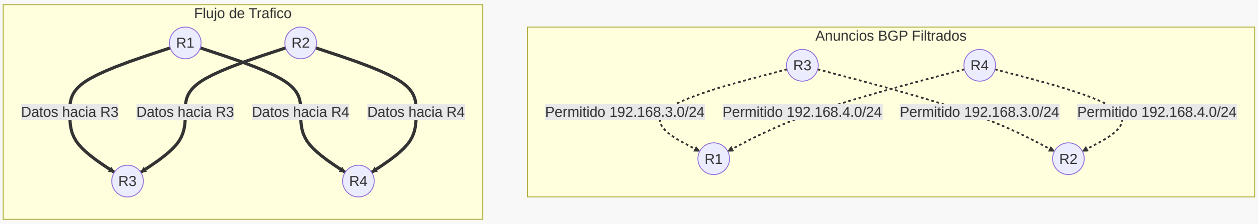
R4

```
R4_Brillantes67(config)# ip prefix-list salida_R4 seq 10 permit
192.168.4.0/24
R4_Brillantes67(config)# router bgp 65004
R4_Brillantes67(config-router)# neighbor 192.168.10.1 prefix-list salida_R4
out
R4_Brillantes67(config-router)# neighbor 192.168.10.2 prefix-list salida_R4
out
```

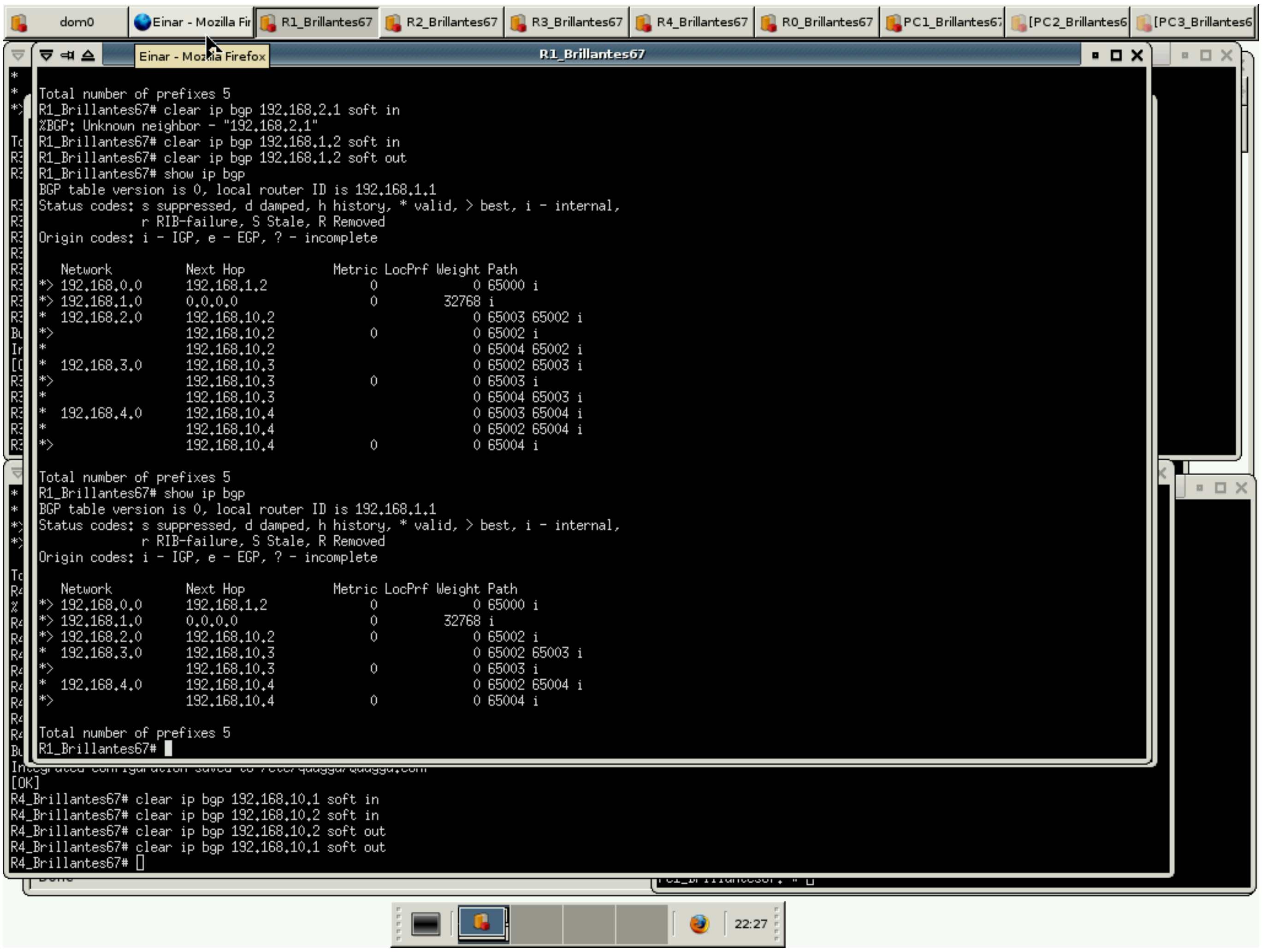
R3

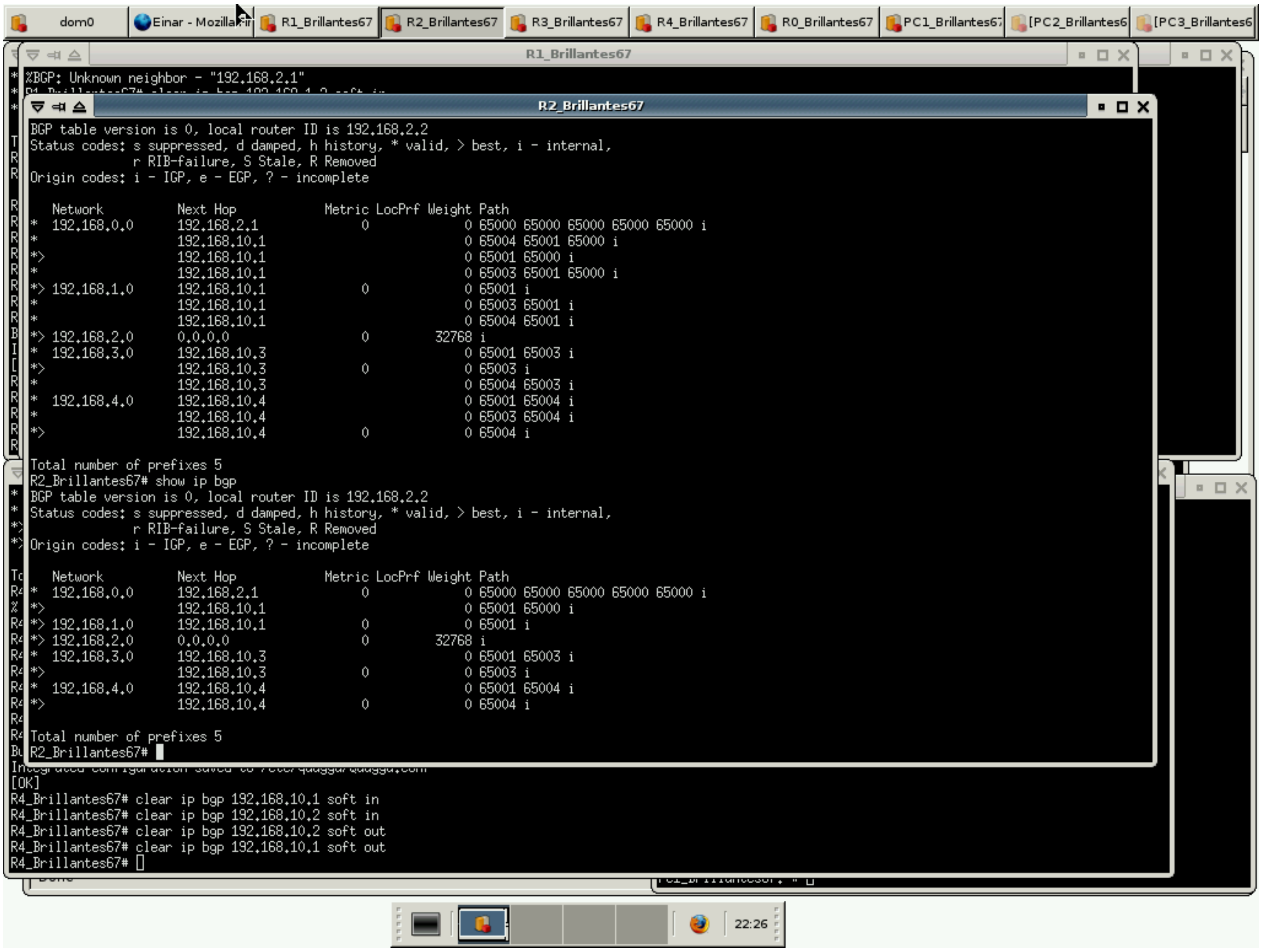
```
R3_Brillantes67(config)# ip prefix-list salida_R3 seq 10 permit
192.168.3.0/24
R3_Brillantes67(config)# router bgp 65003
R3_Brillantes67(config-router)# neighbor 192.168.10.1 prefix-list salida_R3
out
```

```
R3_Brillantes67(config-router)# neighbor 192.168.10.2 prefix-list salida_R3
out
```

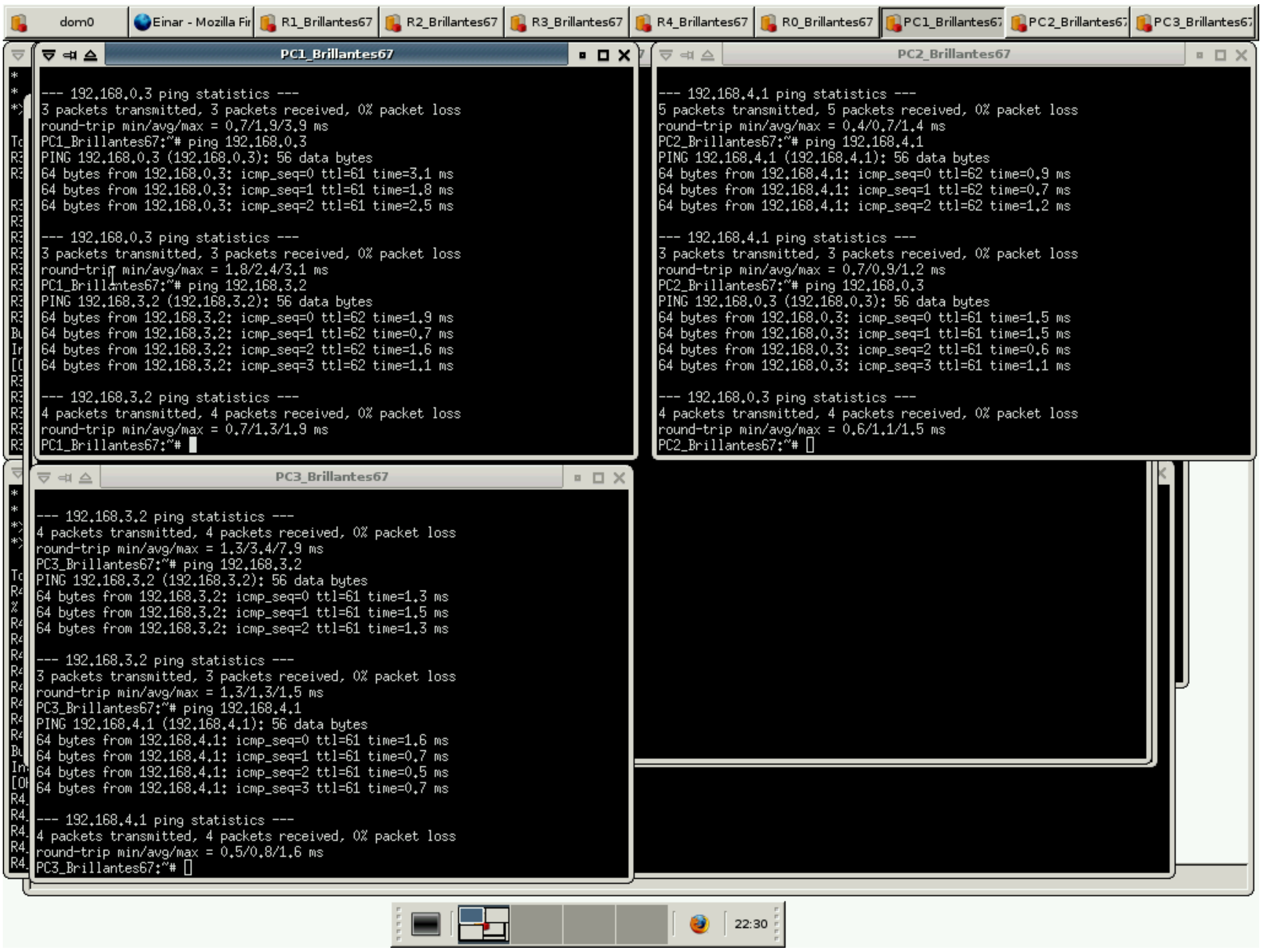


Esto lo que hace es permitir que solo la red 3.0/24 sea visible en la salida para R1 y R2 en el caso de R3 y lo mismo para R4 por lo que si tiene que pasar informacion a R3 desde R1 o R2 solo lo hara si se debe llegar a 3.0/24, antes tambien dejaba usar a R4. Esto se ve en las imagenes.


dom0
Einar - Mozilla Fir
R1_Brillantes67
R2_Brillantes67
R3_Brillantes67
R4_Brillantes67
R0_Brillantes67
PC1_Brillantes67
[PC2_Brillantes6
[PC3_Brillantes6
Einar - Mozilla Firefox
R1_Brillantes67
Total number of prefixes 5
R1_Brillantes67# clear ip bgp 192.168.2.1 soft in
%BGP: Unknown neighbor - "192.168.2.1"
R1_Brillantes67# clear ip bgp 192.168.1.2 soft in
R1_Brillantes67# clear ip bgp 192.168.1.2 soft out
R1_Brillantes67# show ip bgp
BGP table version is 0, local router ID is 192.168.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
Network Next Hop Metric LocPrf Weight Path
*> 192.168.0.0 192.168.1.2 0 0 65000 i
*> 192.168.1.0 0.0.0.0 0 32768 i
* 192.168.2.0 192.168.10.2 0 65003 65002 i
*> 192.168.10.2 0 0 65002 i
* 192.168.10.2 0 65004 65002 i
* 192.168.3.0 192.168.10.3 0 65002 65003 i
*> 192.168.10.3 0 0 65003 i
* 192.168.10.3 0 65004 65003 i
* 192.168.4.0 192.168.10.4 0 65003 65004 i
* 192.168.10.4 0 65002 65004 i
*> 192.168.10.4 0 0 65004 i
Total number of prefixes 5
R1_Brillantes67# show ip bgp
BGP table version is 0, local router ID is 192.168.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
Network Next Hop Metric LocPrf Weight Path
*> 192.168.0.0 192.168.1.2 0 0 65000 i
*> 192.168.1.0 0.0.0.0 0 32768 i
*> 192.168.2.0 192.168.10.2 0 0 65002 i
* 192.168.3.0 192.168.10.3 0 65002 65003 i
*> 192.168.10.3 0 0 65003 i
* 192.168.4.0 192.168.10.4 0 65002 65004 i
*> 192.168.10.4 0 0 65004 i
Total number of prefixes 5
R1_Brillantes67#
[OK]
R4_Brillantes67# clear ip bgp 192.168.10.1 soft in
R4_Brillantes67# clear ip bgp 192.168.10.2 soft in
R4_Brillantes67# clear ip bgp 192.168.10.2 soft out
R4_Brillantes67# clear ip bgp 192.168.10.1 soft out
R4_Brillantes67#
22:27

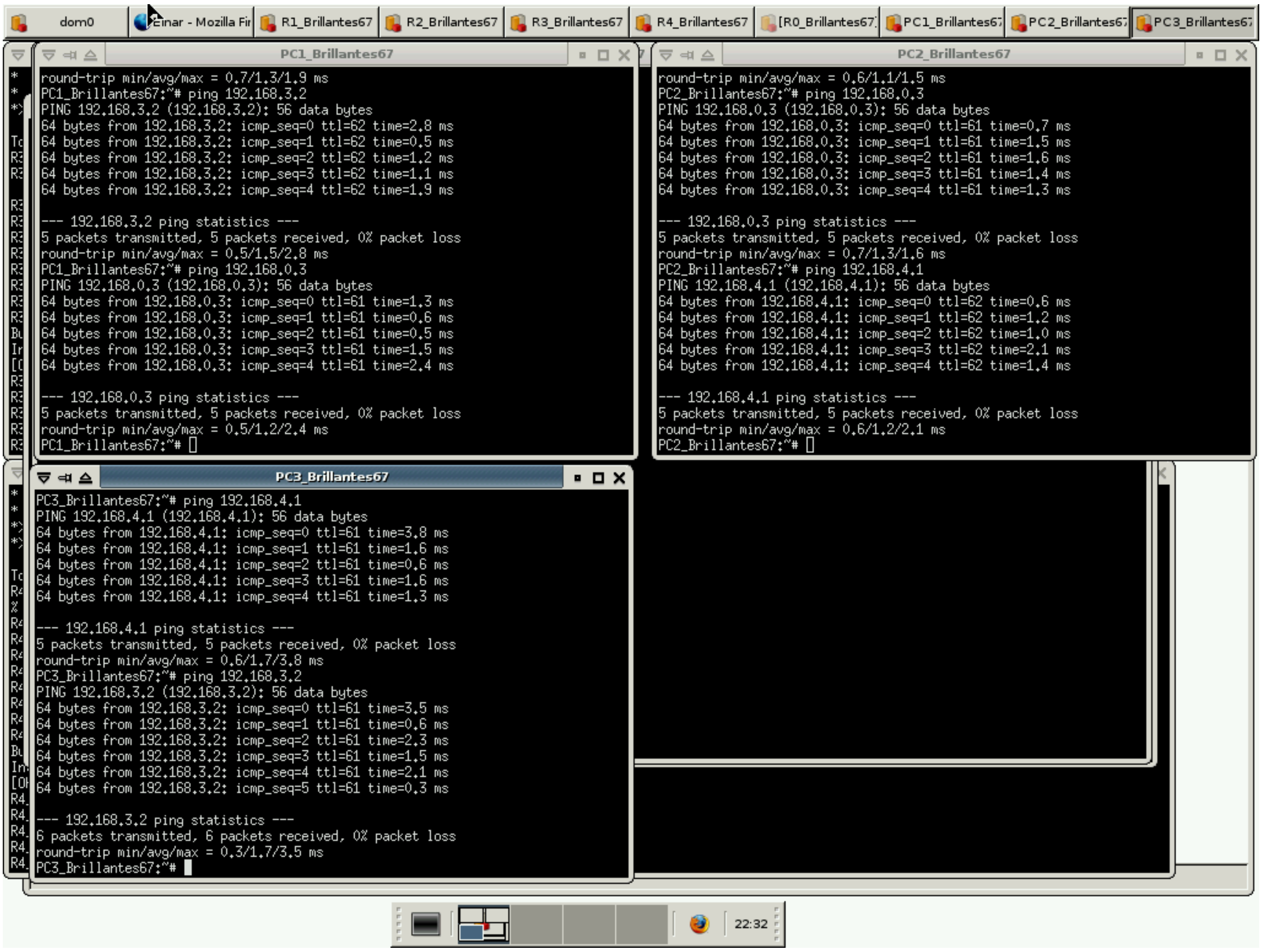


Finalmente las conexiones entre todos los pares de computadores funcionan.



8-Shutdown

Las conexiones siguen funcionando. Ahora solo toman la ruta por R2 para llegar a donde deberian. Se nota que el tiempo promedio (aunque son pocos paquetes) sube un poco y puede ser debido a lo hecho en el paso 6.


dom0
Einar - Mozilla Fir
R1_Brillantes67
R2_Brillantes67
R3_Brillantes67
R4_Brillantes67
[R0_Brillantes67]
PC1_Brillantes67
PC2_Brillantes67
PC3_Brillantes67
PC1_Brillantes67
round-trip min/avg/max = 0.7/1.3/1.9 ms
PC1_Brillantes67:~# ping 192.168.3.2
PING 192.168.3.2 (192.168.3.2): 56 data bytes
64 bytes from 192.168.3.2: icmp_seq=0 ttl=62 time=2.8 ms
64 bytes from 192.168.3.2: icmp_seq=1 ttl=62 time=0.5 ms
64 bytes from 192.168.3.2: icmp_seq=2 ttl=62 time=1.2 ms
64 bytes from 192.168.3.2: icmp_seq=3 ttl=62 time=1.1 ms
64 bytes from 192.168.3.2: icmp_seq=4 ttl=62 time=1.9 ms
--- 192.168.3.2 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.5/1.5/2.8 ms
PC1_Brillantes67:~# ping 192.168.0.3
PING 192.168.0.3 (192.168.0.3): 56 data bytes
64 bytes from 192.168.0.3: icmp_seq=0 ttl=61 time=1.3 ms
64 bytes from 192.168.0.3: icmp_seq=1 ttl=61 time=0.6 ms
64 bytes from 192.168.0.3: icmp_seq=2 ttl=61 time=0.5 ms
64 bytes from 192.168.0.3: icmp_seq=3 ttl=61 time=1.5 ms
64 bytes from 192.168.0.3: icmp_seq=4 ttl=61 time=2.4 ms
--- 192.168.0.3 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.5/1.2/2.4 ms
PC1_Brillantes67:~#
PC2_Brillantes67
round-trip min/avg/max = 0.6/1.1/1.5 ms
PC2_Brillantes67:~# ping 192.168.0.3
PING 192.168.0.3 (192.168.0.3): 56 data bytes
64 bytes from 192.168.0.3: icmp_seq=0 ttl=61 time=0.7 ms
64 bytes from 192.168.0.3: icmp_seq=1 ttl=61 time=1.5 ms
64 bytes from 192.168.0.3: icmp_seq=2 ttl=61 time=1.6 ms
64 bytes from 192.168.0.3: icmp_seq=3 ttl=61 time=1.4 ms
64 bytes from 192.168.0.3: icmp_seq=4 ttl=61 time=1.3 ms
--- 192.168.0.3 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.7/1.3/1.6 ms
PC2_Brillantes67:~# ping 192.168.4.1
PING 192.168.4.1 (192.168.4.1): 56 data bytes
64 bytes from 192.168.4.1: icmp_seq=0 ttl=62 time=0.6 ms
64 bytes from 192.168.4.1: icmp_seq=1 ttl=62 time=1.2 ms
64 bytes from 192.168.4.1: icmp_seq=2 ttl=62 time=1.0 ms
64 bytes from 192.168.4.1: icmp_seq=3 ttl=62 time=2.1 ms
64 bytes from 192.168.4.1: icmp_seq=4 ttl=62 time=1.4 ms
--- 192.168.4.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.6/1.2/2.1 ms
PC2_Brillantes67:~#
PC3_Brillantes67
PC3_Brillantes67:~# ping 192.168.4.1
PING 192.168.4.1 (192.168.4.1): 56 data bytes
64 bytes from 192.168.4.1: icmp_seq=0 ttl=61 time=3.8 ms
64 bytes from 192.168.4.1: icmp_seq=1 ttl=61 time=1.6 ms
64 bytes from 192.168.4.1: icmp_seq=2 ttl=61 time=0.6 ms
64 bytes from 192.168.4.1: icmp_seq=3 ttl=61 time=1.6 ms
64 bytes from 192.168.4.1: icmp_seq=4 ttl=61 time=1.3 ms
--- 192.168.4.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.6/1.7/3.8 ms
PC3_Brillantes67:~# ping 192.168.3.2
PING 192.168.3.2 (192.168.3.2): 56 data bytes
64 bytes from 192.168.3.2: icmp_seq=0 ttl=61 time=3.5 ms
64 bytes from 192.168.3.2: icmp_seq=1 ttl=61 time=0.6 ms
64 bytes from 192.168.3.2: icmp_seq=2 ttl=61 time=2.3 ms
64 bytes from 192.168.3.2: icmp_seq=3 ttl=61 time=1.5 ms
64 bytes from 192.168.3.2: icmp_seq=4 ttl=61 time=2.1 ms
64 bytes from 192.168.3.2: icmp_seq=5 ttl=61 time=0.3 ms
--- 192.168.3.2 ping statistics ---
6 packets transmitted, 6 packets received, 0% packet loss
round-trip min/avg/max = 0.3/1.7/3.5 ms
PC3_Brillantes67:~#
22:32